# 9.8 Nobody Gets Their Own Shirt

**WORKSHEET**

**The question**

**STEP II 2016, Question 12**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q12

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

**Pause and try** Try $n = 3$: list all six orders of cards $1, 2, 3$ and count how many have no card in its own place. Then try $n = 4$, if you are patient (there are 24 orders).

### STAGE 1 Inclusion–exclusion for three events

**Ask yourself** How can you write $A \cup B \cup C$ as a union of two events, so that the two-event formula applies?

### STAGE 2 The probabilities

**Ask yourself** If card $i$ is in position $i$, how many ways are there to arrange the rest? And if cards $i$ and $j$ are both fixed?

### STAGE 3 Adding it up

**Ask yourself** In the inclusion–exclusion formula for $\mathrm{P}(E_1 \cup \cdots \cup E_n)$, how many terms are there involving intersections of $k$ events, and what is their total?

### STAGE 4 Exactly one

**Ask yourself** For exactly one card to be in place, which card is it, and what must the others do?

## Try this next

1 **Shirts.** Try STEP III 2004, Question 12 (step.maths.org/questions/04-s3-q12). It counts the players wearing their own shirts with indicator variables, finds the mean and variance, and compares the result with a Poisson distribution.

2 **A recurrence.** Let $D_n$ be the number of derangements of $n$ objects. By considering where card 1 goes, and whether the card in that position goes to position 1, show that $D_n = (n-1)(D_{n-1} + D_{n-2})$. Check it against $D_3 = 2$, $D_4 = 9$.

3 **Exactly $k$.** Show that the probability that exactly $k$ cards are in their own places is $\frac{1}{k!} d_{n-k}$, and that it tends to $\frac{e^{-1}}{k!}$.

4 **Coprime integers.** Use inclusion–exclusion to count the integers from 1 to 30 divisible by none of 2, 3 and 5. How does your answer relate to $30(1-\frac{1}{2})(1-\frac{1}{3})(1-\frac{1}{5})$?

## Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press). A first-year text that covers inclusion–exclusion and the matching problem.
- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter IV. A classic, with many variations on the matching problem.
- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), on inclusion–exclusion.

The full guided solution, and what the question leads to at university, is in Section 9.8:
https://beyondstep.keithmcnulty.org/ch09/derangements.html